Expert Systems with Applications 159 (2020) 113642

Contents lists available at ScienceDirect

Expert Systems with Applications

journal homepage: www.elsevier.com/locate/eswa

Review

Influential nodes detection in dynamic social networks: A survey

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ARTICLE INFO

Article history:
Received 12 November 2019
Revised 5 March 2020
Accepted 5 June 2020
Available online 13 June 2020

Keywords:
Influential nodes
Influence maximization
Dynamic social networks

ABSTRACT

The influence maximization problem has gained increasing attention in recent years. Previous research focuses on the development of algorithms to analyze static social networks. However, real social networks are not static but they are represented as dynamic networks that evolve across time. Motivated by this drawback, the purpose of this survey is to highlight the characteristics and challenges of the influential nodes detection problem. A classification of published approaches should be proposed. This work is organizing state-of-the-art methods into a technical comparison that are based on network models. Due to the definition of network models and the influential nodes detection problem, this survey will help researchers to find the set of methods best suited for their needs. The proposed classification could also help researchers to select the right direction in which their future research should be oriented.

1. Introduction

A social network plays an essential role in the spread of influence. Identifying a group of important nodes in a social network is a fundamental issue, so that they are initially targeted (e.g., to adopt a new product) and that can maximize the adoption of the new product. Social networks are classified as static and dynamic social networks. It is relatively simple to identify influential nodes in static networks. To accomplish this, many algorithms are proposed. Recently, it has become more important to detect influential nodes in a dynamic social network. As the network grows in size, the formation of new nodes takes place and previous nodes dissolve. Existing researches and solutions on influential nodes detection focus mainly on developing efficient and effective algorithms on a static social network. However, researchers focus more on the social networks' dynamic aspect.

One of the most challenging problems is the influential node identification in dynamic social networks that has gained increasing attention in recent years. Real social networks like Facebook or Twitter are highly changeable therefore they are represented as temporal graphs that evolve over time. Temporal graphs model real networks that exhibit a dynamic behavior where the interactions between elements of the network change over time. As an example, we can consider an online social network where each day friends are added and others are deleted over time or a mobile communications network where the connections represent calls between mobile phones.

Social networks are dynamic in the actual world and change over time. Users create their relations in social networks, while user interactions vary over time. A temporal network can be modeled as a series of static networks. As it is already published in existing research, we assume that there are three types of operations to apply on dynamic social networks. Due to advances in data collection technologies, it is becoming increasingly common to study dynamic networks. An important research issue is how to identify influential nodes in dynamic social networks.

Operations on dynamic social networks are more complex. Palla, Barabási, and Vicsek (2007), which listed six different operations, first introduced them. Seven operations are identified in Cazabet, Takeda, Hamasaki, and Amblard (2012). In this review, the evolution of the network is performed using nine different operations: operations are applied on nodes where new nodes, added to or removed from old nodes, operations are applied to the edges where a connection is removed or created between two nodes and operations are applied to the communities.

A list of these operations is presented in Table 1. To reduce the amount of computation, Liu et al. (2017) proposed a method to efficiently quantify changes in the spread of node influences. The method supports the concept of localization and aims to limit the spread of influence to the node regions. The principal concept of localization is to use each node's local region to estimate its overall influence distribution. During the diffusion process, they

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https://doi.org/10.1016/j.eswa.2020.113642
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Table 1
Description of the operations in the context of a social network.

Operations	Description
Add node	A new actor joins the network in an existing community
Remove node	An actor leaves the network
Add within edge	A connection is created between two actors belonging to the same community
Remove within edge	A connection is removed between two actors belonging to the same community
Add between edge	A connection is created between two actors in distinct communities, i.e., creation of a bridge between two communities
Remove between edge	A connection is removed between two actors in distinct communities
Split community	An existing community is split into two new communities, and connections are removed between distant members of the original community
Merge community	Two similar communities are merged into a single one, and new connections are created between the members of the new community
Migrate community	A group of actors leave their community to either create a new one or join an existing one

consider only paths (*addEdge*) whose probability of influence is greater than θ while neglecting those with a probability lower than θ. In doing so, the influence is limited to each node's local region. The authors suggest that *addNode* (or *removeNode*) is the most straightforward operation since it simply sets the node's influence spread to 1 (or 0); *addWeight* and *decWeight*, as well as *removeEdge*, are methodologically similar to *addEdge*. In Wang, Zhu, and Ming (2017), the authors described only the operation of AddEdge and DelEdge, because the other operations were identical. In the AddEdge algorithm, if the probability of the new path from a node u to v is higher than threshold θ, then this path is added to *temp* (used to temporarily store the new path). In DelEdge algorithm, they validate all paths of node u that is impacted by the update, and calculate the influence spread change of node u and delete all invalid paths. In Wang, Cuomo, Mei, Cheng, and Xu (2019), the authors build a dynamic evolution model to generate a series of dynamic networks by adding and/or removing vertices and/or edges from the network obtained in the previous time step. By repeating the process (Fig. 1) a series of complex networks at different time steps is created. The input of the initial network to the dynamic evolution model process is an edge list; then according to the edge list, the vertex list is generated.

Most papers that studied the community tracking problem focusing on the set of simple operations that involve entities of a dynamic network: node/edge adding and removing. In Boudebza, Cazabet, Azouaou, and Nouali (2018), the authors proposed OLCPM: An Online Framework for Detecting Overlapping Communities in Dynamic Social Networks. This framework is based on analyzing the network's dynamic behaviours, that can result from the insertion or deletion of nodes or edges. Every time a change is created in the network, the community structure is modified locally across the modified nodes. The evolving node or edge technique helps authors to identify the growth, split, death and merge of communities. In Zhao, Li, Zhang, Chiclana, and Viedma (2019), the main idea was to identify communities in the initial state, then collect and analyze incremental dynamic changes and gradually update communities. The incremental elements ΔG^{t+1} are applied to all changes between time t and $t+1$, which might be composed of several subgraphs. The incremental subgraphs strategy helps authors to identify the birth of a new community, the enlarging of the previous community, the death and division of an old community. Research in the area of dynamic social network has become very crucial in order to solve the problem of influential nodes detection. Many methods have been presented. The remaining paper will throw a light on different algorithms that were proposed for influential nodes detection in dynamic social networks.

In this review, to classify the influential nodes detection algorithms, we consider the technical solution that the authors use to find the influential nodes, but rather we study the model they use to identify them. Our goal is to support researchers in discovering the approach that suits their needs and data, or at least the category of approaches. For each proposed class of influential node detection algorithms, we discuss both advantages and drawbacks. Moreover, we provide a brief description of those methods and of network models.

The review is structured in the following manner. The general concepts behind the network models are discussed in Section 2. In Section 3, we define the influential nodes detection problem. In Section 4, we describe our classification of the different influential nodes detection proposed algorithms. In Section 5, we compare the different classes of algorithms. In Section 6, we discuss some challenges and future directions of the influence maximization problem. Finally Section 7 concludes the review, providing insights into influential nodes detection related open issues.

2. Network models

Time plays a crucial role in reconstructing network topologies. Fig. 2 shows three main conceptual solutions that gradually introduce the temporal dimension in the modeling process of the network.

As Rossetti (2015) introduced, switching from static to dynamic models increases the complexity of the model as the grain of the available temporal information decreases.

2.1. Static networks

Social networks have been used to explain and understand a variety of problems. A research issue in Krackhardt and Porter (1985) in micro organizational behavior is analyzed with the help

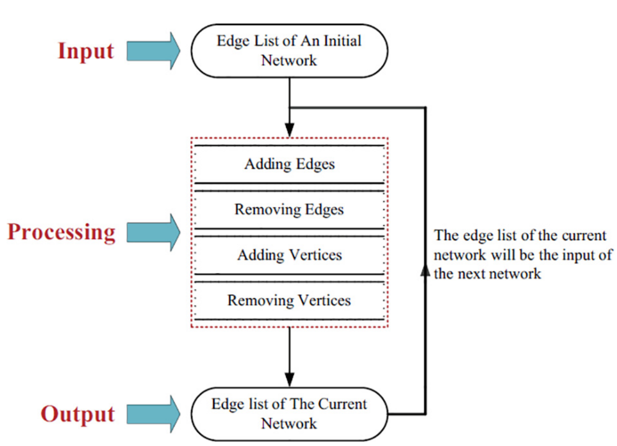

Fig. 1. Process of the dynamic evolution model.

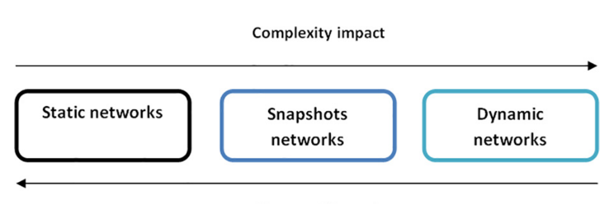

Fig. 2. Network representations.

of theories in psychology and sociology, job-related rewards (Bian, 1997). In Bian (1997) and based on Granovetter (1995), the authors observe that in China social networks are exploited to influence authorities who consecutively attribute jobs as favors to their contacts that is a style of unauthorized activity facilitated by strong ties defined by power and obligation. In Brass (1984), influence and power are studied, and in Labianca, Brass, and Gray (1998) a study is carried out within an organization to examine the relationship between members of different departments and the perception of intergroup conflicts.

Previous researchers have studied social networks from a static perspective while ignoring the perspective of dynamic evolution of social networks over time (Marsden, 1990).

A static social network looks like a graph structure consisting of nodes and edges, where the nodes represent a social entity and the edges represent the links, in other words, the interactions between the connected nodes. A static social network can be modeled by a graph $G = \{V, E\}$ which consists of a set of nodes V and a set of edges E connecting them.

2.2. Snapshots networks

To compute activation probabilities of nodes in social networks, Kossinets and Watts (2006); Backstrom, Huttenlocher, Kleinberg, and Lan (2006) and Shi, Zhu, Cai, and Zhang (2009) used the snapshot definition though with widely varying numbers of snapshots.

A snapshot network is a static network corresponding to alive nodes and edges at a given time in dynamic social networks. The social network can be divided into a series of snapshots where each of them is used to describe the network status at a time t (network relationship) or to aggregate interactions observed over a period of time (network interactions). In many proposed approaches, the social network is modeled as a series of snapshot graphs $\{G1, G2, \ldots, Gn\}$ where the edges and the nodes in each snapshot graph change across different time intervals. We denote the snapshot graph as $G^t = (V^t, E^t)$, where $V^t = \{v_1, \ldots, v_i, \ldots, v_n\}$ represents nodes and E^t represents edges appearing during the periods under consideration. Notation $G^t, t = 0, \ldots, T$ defines the snapshot graphs over time. An example of a social network that contains four snapshots is illustrated in Fig. 3.

2.3. Dynamic networks

Various formalisms are suggested to reflect changing networks without loss of information: Temporal Networks (Holme & Saramäki, 2012), Time-Varying Graphs (Casteigts, Flocchini, Quattrociocchi, & Santoro, 2012), Interaction Networks Rossetti, 2015, Link Streams and Stream graphs (Latapy, Viard, & Magnien, 2018). In this review, we use the term Dynamic networks.

An interesting perspective is shown in Perry-Smith and Shalley (2003) offering a better understanding of the importance of social networks. The authors suggest a factor, creativity, that explains how and when the position of an individual in the network can change from one position to another. This is important because it presents a more dynamic and fluid image of social networks and responds to some of the criticisms of social network research that networks are static and not addressing the evolution of the network (i.e. Marsden, 1990).

In Holme and Saramäki (2012), the authors present the emergent field of temporal networks and discuss methods for analyzing the topological and the temporal structure and models for illustrating their link with the dynamic behavior of social networks.

A dynamic network is designed as a dynamic structure in which nodes and edges can be added or removed from the network over time. A dynamic social network contains not only a set of relationships between nodes but also information on how these relationships change over time. Fig. 4 shows a network's dynamic structure at different timestamps. We take as an example the social network, where each node indicates a user, each edge represents a connection between two users, and the weight associated with each edge shows the energy that users spend to establish their connection. User D and user E have similar structures at timestamp t: each of them has three friends who are unknown. Therefore, both of them invest most energy on a best friend (i.e. spending 0.8 energy on a best friend while spending 0.2 energy overall on others). Thus, traditional network methods can construct identical representation vectors for two users A and B. However, their patterns of evolution are distinct. For example, two pairs of D's friends connect to each other at timestamp $t+1$, while there are still no ties between E's friends. Additionally, user E spends its energy at timestamp $t+1$ on other friends, while user D continues to focus on developing its relationship with similar friends from the timestamp t. Various user patterns illustrate the differences between their own characters and social strategies that are used as connections to others to meet their social needs. For example, user D tends to introduce its friends to each other, while user E prefers to keep their friends in their own communities close. Therefore, a crucial criterion for network embedding methods is whether the learned representations can represent the evolution patterns of vertices. Network embedding is a powerful form of network representation, can support subsequent network processing and analysis tasks such as node classification, node clustering, network visualization and link prediction (Cui, Wang, Pei, & Zhu, 2018). The traditional topology-based network representation usually uses the observed adjacency matrix which can sometimes include noise or redundant information. The main objective of the embedding-based representation is to study the dense and continuous representations of nodes in a low-dimensional space, so that the noise or redundant information can be reduced and the intrinsic structure information can be preserved. Network embedding

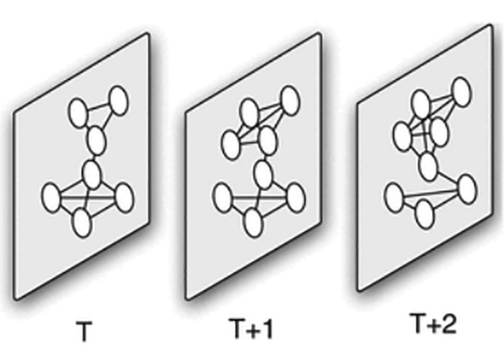

Fig. 3. Snapshot graph.

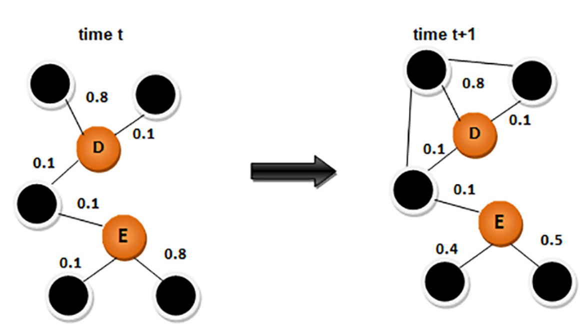

Fig. 4. An illustration of dynamic social network.

will open opportunities for network analyzes to benefit from the rich literature on machine learning. Some off-the-shelf machine learning approaches, like deep learning models, can be applied directly to address network problems.

3. Influence maximization problem

The influence is the ability of a user to lead another user to adopt an idea, an opinion, to believe in information, to use services or products.

When diffusing an idea or an information in social networks, the main objective is to extract a set of users that maximized the influence within a network. The main question is: how to maximize the influence within a social network?.

Kempe, Kleinberg, and Tardos (2003) are the first who formulate the influence maximization problem (IM) as an optimization problem which consists in finding the K best initial users (called seeds) that maximize the spread of influence across the social network. The classical problem of Influence Maximization aims to find influential nodes for just one static social network. However, real-world networks are dynamic. The structure and the influence value associated with the edges are constantly changing. As a result, the set of nodes that increases the influence degree must be constantly updated as the network structure and the influence value evolve.

3.1. Key concepts

This section is dedicated to define some key concepts used in this paper.

3.1.1. Influence propagation and diffusion models

Social media is one of the best ways for people to find and interact with others. In sociology, many researchers have been done on this field, through studies on effects such as opinion formation and the diffusion of innovations (Rogers, 1995). In economics (Strang & Soule, 1998), theoretical models have been proposed to make social influence a process by which members of a social network tend to coordinate (or prevent coordination) their decisions (Blume, 1993; Young, 2001). More recently, researchers have developed models of influence on social networks, driven by applications such as viral marketing (Domingos & Richardson, 2001; Kempe et al., 2003; Leskovec, Adamic, & Huberman, 2007), online news dissemination (Gruhl, Guha, Liben-Nowell, & Tomkins, 2004; Leskovec, McGlohon, Faloutsos, Glance, & Hurst, 2008).

Even if we consider network structures as static objects that do not allow edges and nodes to be created and removed, there are interesting issues that are closely related to temporal analysis. One of the most popular issues is the spread of information. Users in a social network interact (i.e., they can post on their wall on Facebook, express their support through likes and tags on images and comments of their friends), exchange information (i.e., in a professional network news about openings of positions) check and promote topics and trends (i.e., hashtags on Twitter). Many of these actions performed by users on a social network can produce cascading phenomena: the aim of studies that address diffusive phenomena is to understand how they contribute to social influence, how knowledge spreads to friends recursively from a user, and how this process could create tribes or groups.

The study of diffusion in social networks is mainly done through the development of models that try to best represent the diffusion process. Influence propagation models in social networks try to reproduce the diffusion of information between users. These models are used in many applications, such as identifying influence nodes or selecting initial seeds to maximize the influence propagation process.

There are two main used models in the literature and they admit two states of nodes: active and inactive. The first model is the Linear Threshold model (LT) introduced by Granovetter (1978), in which each v node has a θ_v threshold. At a time t, the parent nodes of v that are active are trying to activate it, v will only be active if the sum of $W_{u,v}$ is greater than the threshold θ_v (Fig. 5).

The second type of model is the Independent Cascades model (IC) introduced by Goldenberg, Libai, and Muller (2001) where as soon as a node u is active, It has a unique opportunity to activate each of its direct neighbors v, with a probability Pu, v (Fig. 6).

Several other models have been proposed like the Weighted Cascade (WC) introduced in Wang and Yang (2009), if a node u is activated in round i, with a probability 1/dv, v is activated by u in round $i+1$. As with the IC model, each neighbor is able to activate v independently. Therefore, if a not yet activated node v has l neighbors disabled during the i-th round, the probability that v will be activated in round $i+1$ is $1-(1-1/dv)$.

Kempe et al. (2003) proposed the triggering model (TR) that evolves as follows: Each vertex I selects a random set of its neighbors as triggers, where the for a given node th choice of triggers is independent of the choice for all other nodes. If a node i becomes uninfected at the time t, however a node in its set of triggers becomes infected, node i becomes infected at time $t+1$.

Ren, Wang, Liu, Wang, and Dong (2016) introduced the Susceptible Infectious Recovered model (SIR), there are two compartments: (1) Susceptible S(t) represents the number of susceptible individuals (not yet infected) to the disease, (2) Infected I(t) denotes the number of people infected and are able to propagate the disease to susceptible. people. For the first time, a node is selected to be infected. Each node in the infected state selects its susceptible neighbors randomly with probability, and remains infected. The process of spreading stops when no infected node is present.

3.1.2. Metrics

Over the years, various centrality metrics have been proposed and applied in network analysis to estimate node importance and to characterize the role or the position of influential nodes in social networks. Such basic centrality metrics are degree centrality, betweenness centrality, and closeness centrality.

- Traditionally first and conceptually the easiest node metric is degree centrality, where a degree of a node is the number of edges the given node has Wasserman et al. (1994). The Degree Centrality consists of measuring the sum of the intensity of the connection of a node with its direct neighbors.
- The betweenness centrality is a measure extent to which a node is attached indirectly to other nodes of the social network via its direct links. It is an important technique in social network analysis, especially in the community detection problem. However, this technique suffers from a high execution time. Like introduced in Brandes (2001), the best algorithm for the betweenness centrality requires a time complexity equivalent to O (NM) for unweighted networks with N nodes and M edges.
- The Closeness Centrality is the degree to which a node is near to all other nodes in a social network. The closeness centrality rates are high for individuals closer to the center of local clusters. Therefore, the closeness measure overallocates influential users next to each other (Morone & Makse, 2015). Moreover, closeness centrality is unsuitable for large-scale social networks due to its high time complexity (Chen, Lü, Shang, Zhang, & Zhou, 2012).

Another metric that is widely applied in the literature to analyze relationships in social networks is the semantic similarity

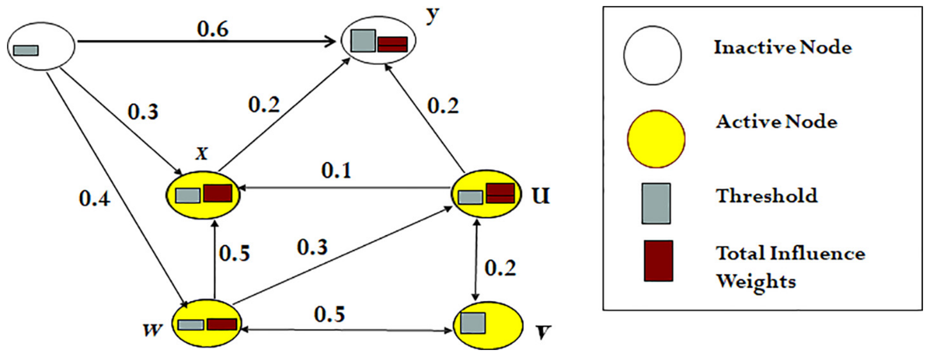

Fig. 5. LT model.

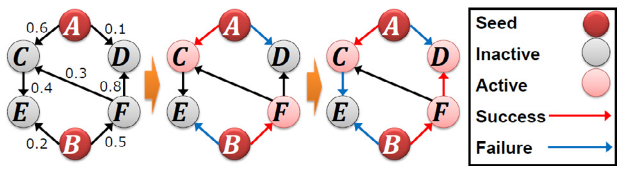

Fig. 6. IC model.

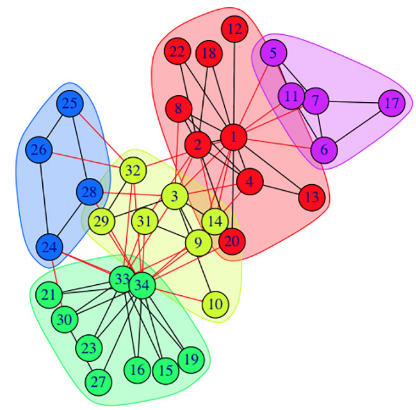

Fig. 7. Community detection problem.

such as that of Correspondence, Jaccard and Cosine. It consists in extracting the common elements between two nodes.

However, all of these metrics are applied to static networks and they do not consider the feature that real social networks are dynamic. Then, it's illogical to use them directly in dynamic networks since they do not contain information about the activity time of nodes. Motivated by this deficiency, many approaches that study the problem of influential nodes detection in dynamic social network have been proposed.

3.1.3. Influence maximization and Community detection problems

Community detection is one among the most popular issues in the field of social network analysis. Considering the paper (Girvan & Newman, 2002), different approaches have been proposed on the topic. From the initial problem of graph partitioning, in which each network node should belong to one and only one community, new issues related to the structural aspect of the community were taken into consideration have been studied, such as overlapping communities and hierarchical decomposition. Recently, new approaches have been proposed, that can manage dynamic networks.

The field of social network analysis can contain multiple areas of research, such as Influence maximization and Community detection. A community is a group of individuals who are interconnected. An important feature of social networks is the presence of community structures, that is, groups of nodes that have a high density of edges and a lower density between them. The community detection problem consists in finding, for a given graph, an optimal partitioning in k partitions (Fig. 7).

By studying the approaches that have been proposed in these two fields, we find that they are connected by a particular set of actors who played an important position in the network. These actors are known as influencers, whether for the survival of the community or the spread of influence in a network, it is still necessary that some actors maximize the diffusion of influence over the rest of the community. These actors assign to a special community of the individual. So, instead of targeting the entire network to find influential nodes, you can start by finding the community, where the degree of influence is high, and then finding seed nodes. Understanding the network structure evolution is of primary importance to conceive an influence maximization algorithm for dynamic social networks.

Understanding the network topology evolution is of primary importance to design an influence maximization algorithm for dynamic social networks. The fastly and unpredictably evolving topology of the dynamic social networks poses many challenges in the studying of the problem of influential nodes detection, which we list as follows: Firstly, in real-world social networks, the interrelationships between edges are complicated. Thus, one small change in the network structure may affect the influence propagation of a large number of nodes, not to indicate the enormous changes in complex social networks. After the evolution, it is very hard to efficiently enumerate the changes of the influence propagation for all the nodes. Secondly, in large-scale social networks, there are a high number of nodes. Thus, it is a very challenging problem to appropriately limit the set of influential nodes and reduce the computation of those nodes to the maximum.

In the community detection problem, a group of individuals who share the same interests tend to influence each other. For example, a person who shares his or her film preferences with another person would be forced to watch a new film if he or she sees positive feedback from the other. When this happens, community detection algorithms increase the parameter used to represent the common interests of these individuals like node importance measurement which is based on the Degree and Closeness Centrality. Hence, the community detection problem becomes connected to the influence maximization problem in a social network.

The community detection problem has a strong influence on the process of identifying influential nodes. To reduce the time complexity of the influence maximization algorithms, many authors proposed approaches that take into account the community structure of social networks. All approaches that apply a community detection algorithm to detect influential nodes adopt a three-phase approach: (a) applying a particular algorithm to identify the network structure; (b) Identifying seed nodes from different communities according to metrics applied in each community and (c) developing a method to select the influential nodes from the seed node set. Fig. 8 (Pozveh, Zamanifar, & Naghsh-Nilchi, 2017).

When applying the diffusion model based on the community structure, we can identify two types of nodes: the "bridge" node which closely connects other communities and is easy to spread information across communities and "influential" node which is at the core of a community and can quickly propagate information within it (Cazabet et al., 2012).

4. Classification

This section discusses the review of various researches propsed in the field of dynamic social network analysis. In this paper, we revise the different approaches done in the problem of identifying influential spreaders in dynamic social networks. In social networks, discovering people who can distribute information to a greater number of people (influential nodes) makes it possible to efficiently propagate high quality of information or new ideas. Several methods on IM can be generally classified as static and dynamic based on their abilities to study the problem of influential nodes detection. In this paper, we summarize them separately as depicted in Fig. 9.

In comparison with other related surveys (Li, Fan, Wang, & Tan, 2018; Bian, Koh, Dobbie, & Divoli, 2019), our survey includes not only research on influence maximization in static social networks but also studies the research area of identifying influential nodes in three network models such as static networks, snapshot networks, and dynamic networks. We focus on introducing the technical aspects of the published approaches to the influence maximization problem in dynamic social networks.

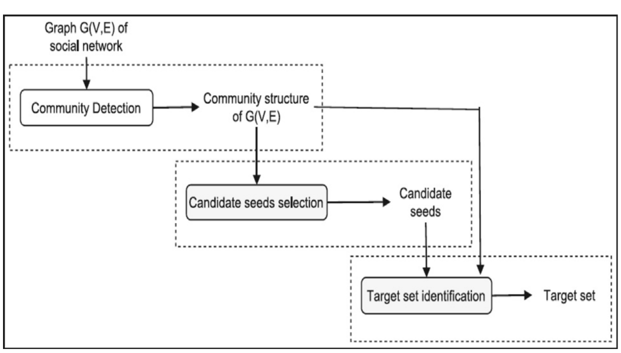

Fig. 8. Community detection with influence maximization.

4.1. Approaches in static networks

This section discusses the review of various researches done in static social network. In Hafiene and Karoui (2017), we have classified the problem of influential nodes detection in static social network into three approaches namely: greedy algorithms, heuristic algorithms and hybrid algorithms.

4.1.1. Heuristic approaches

To study the influential nodes detection problem, many algorithms that apply various heuristics have been proposed. Wang and Yang (2009) propose DegreeDiscount that runs faster than MixedGreedy. The basic idea of Degree Discount Heuritics is, when we consider an edge $\overline{uv}$, with u in the seed set S and v are being considered. Since u is in the seed set, by taking network effect into consideration, the idea is to discount v's degree by one due to the presence of u in the seed set S. The complexity of DegreeDiscount is $O(k \log n + m)$. Degree Discount's basic assumption is not sufficient, as it treats all nodes without any differences. In Narayanam and Narahari (2010), the authors propose the SPIN algorithm to improve the influence diffusion. It chooses k nodes with the highest Shapley values as the target set, where information propagation is observed as a cooperative game. The authors use the Shapley value to describe the marginal contributions that the nodes make to the diffusion process. Wang, Vasilakos, Jin, and Ma (2015) propose price-performance-ratio PPRank inspired from heuristic scheme. It chooses seeds within a given budget and aims to improve the performance of the diffusion process. The authors characterize each user with two distinct factors: the susceptibility of being influenced (SI) and the power of influence (IP) that reflects the ability to actively influence others and evaluate users' SIs and IPs according to their social interactions. A model based on a convex price-demand curve is used to correctly convert each user's SI into a persuasion cost (PC) representing the cost used to make user adopt a new behavior. Additionally, a new cost-effective selection method is introduced by authors that adopt both price performance ratio (PC-IP ratio) and users's influential power (IP) as a selection criterion taking into consideration the overlapping effect. Bae and Kim (2014) propose a coreness centrality which is a novel measure to estimate the influence propagation of a node by using the neighbor's k-shell indices. Based on the authors, the coreness centrality is a greatly improved indicator for assessing the ability of the information diffused through the network. It is measured

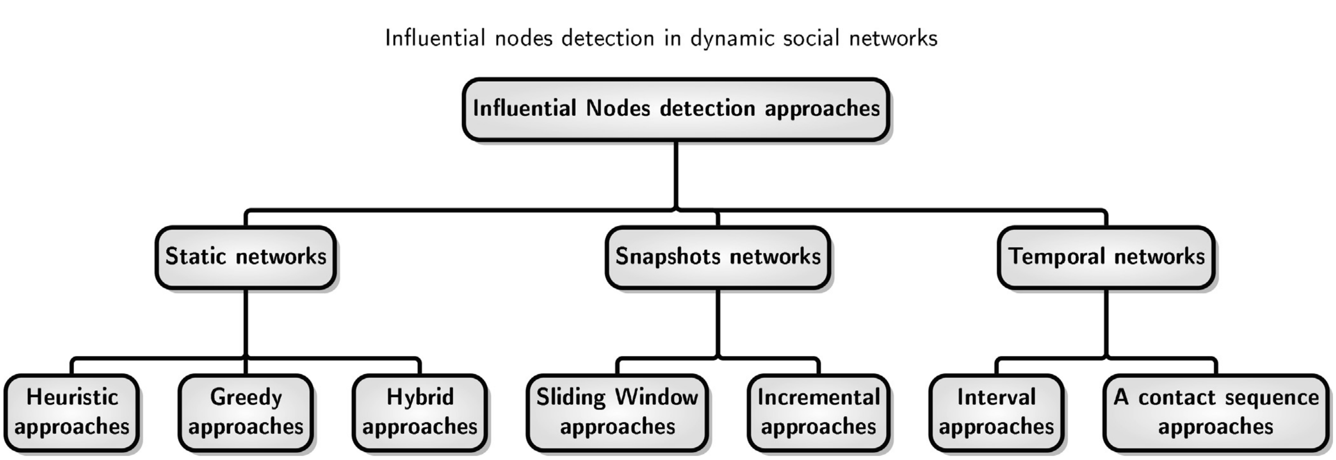

Fig. 9. Approaches of influential nodes detection.

based on the $k-shell$ indices of the adjacent neighbors of a spreader. The technique is based on the idea that a strong spreader providing more connections to the nodes that exist within the center of the network. Thus, K-shell is a powerful measure to evaluate the influence propagation through the network. The approach can consider both the degree and the coreness of a node by counting the k-shell indicators of its neighbor or neighbor of neighbors.

The heuristic approaches use heuristics to study the problem of identifying influential nodes.

Advantages: The most significant advantage of the heuristic approaches is to reduce the number of evaluations on the influence spread of nodes, more scalable on larger networks and faster than the greedy approaches.

Drawbacks: The major problem of heuristic approaches is not suitable for all types of social networks.

4.1.2. Greedy approaches

Greedy approach is designed to provide the best solution for a particular problem. It tries to find an optimal localized solution that could ultimately lead to approximate solutions. Several methods are suggested based on a greedy algorithm. The Greedy's essential idea is to measure each node's influence value and alternately pick the node that maximizes the influence value before selecting K nodes.

CELF (Cost-Effective Lazy Forward) is proposed by Leskovec et al. (2007). In order to identify influential nodes, CELF exploits submodularity. The idea is that the marginal gain of a node in the current iteration cannot be more than that in previous iterations, and thus the number of spread estimations can be significantly reduced. The Triggering Model was introduced by authors to model the diffusion process within a social network. This model generalizes, as the authors demonstrate, the Independent Cascade and the Linear Threshold. CELF is 700 times faster than a simple greedy algorithm. The problem is that CELF demands N spread estimations to establish the initial bounds of marginal increments, which is time expensive on large graphs. Wang and Yang (2009) suggest two algorithms called NewGreedy and MixedGreedy respectively. NewGreedy's basic idea is to remove edges that will not lead to the propagation of the original graph in order to obtain a smaller graph and perform the influence propagation on the smaller graph. MixedGreedy's first step uses the NewGreedy algorithm and uses the CELF algorithm for the remaining steps. They find that MixedGreedy is faster than CELF and NewGreedy while basing on their evaluations. Wang and Yang (2009) apply the function $RanCas(S)$ to denote the random process of influence from the seed set S of which the output is a random set of nodes influenced by S. The main difference between RanCas(S) in the WC model and the IC model is that usually the probability of u activating v is not the same as that of v activating u. Zhou, Zhang, Guo, Zhu, and Guo (2013) proposed an improved CELF algorithm called Lazy Forward (UBLF) based on Upper Bound to classify the important nodes in social networks. In the initialization step of the UBLF algorithm, the authors use the derived upper bound function to further reduce the number of spread estimation. Thus, the nodes will all be ranked in their upper bound scores, which can significantly reduce the computational cost of the original CELF algorithm.

To conclude, one could possibly run any of the static algorithms to optimize the effect of new Top-K nodes. When the social network changes, the runtime of the static algorithms on large scale social network will be extremely long and, whenever the topology of the network changes, we need to recalculate the influence propagation for all nodes, resulting in quite high costs.

The greedy approaches follow the principle of making, step by step, a choice of a local optimum, in the hope of obtaining a global optimum result.

Advantage: The most significant advantage of the greedy approaches is offering good values of influence propagation.

Drawbacks: The major problem of the greedy approaches is not suitable for large networks. However, when the network size increases, the computational time will increase dramatically, which prevents the greedy based algorithm to become a feasible solution for the influence maximization problem in real world.

4.1.3. Hybrid approaches

To study the problem of influential node detection, many approaches suggest a two-phase approach: first, community detection, second, community structure-based detection, and heuristic or greedy approaches identifying influential nodes. A two-phase approach is suggested by Cao, Wu, Wang, and Hu (2011) to combine community detection with greedy algorithms and proposing OASNET (Optimal Allocation in a Social NETwork) to solve the influence maximization problem. OASNET is based on a growth function $F(k, G, M, C)$ that takes four input variables to an expected number of active nodes at the end of the diffusion process, where k is the number of initial active nodes, G is the network, M is the diffusion model, and C is the selection strategy for the initial active nodes. In OASNET, there exists a growth function for each community where they are all disconnected.

The problem is that the growth function doesn't work easily. To approximate that function, the authors need to stimulate the diffusion process on the network many times. In Lv, Guo, and Ren (2013), every social network group is built as a cooperative game player and the Shapley value of that group is represented, as a measure that depended on the density of each community. The number of nodes chosen by each group as target nodes is related to their Shapley value. The nodes are then chosen from two separate node classes in each community: bridge nodes and influential nodes obtained by applying the MixedGreedy (Wang & Yang, 2009).

Rahimkhani, Aleahmad, Rahgozar, and Moeini (2015) suggested a method named ComPath that is based on the linear model to find top-k most influential nodes in social networks. They applied the degree centrality measure to identify the important nodes in the communities. Thus, each community's number of nodes detected is related to its size. The main advantage of ComPath is to reduce the number of identified nodes without loss of quality by: first detecting communities and then investigating a limited number of them to decrease the runtime. Song, Zhou, Wang, and Xie (2014) propose an algorithm called the Community-based Greedy Algorithm (CGA) to detect the influential nodes. It covers two parts: dividing the mobile social network into several communities by taking the diffusion information into account and then applying a dynamic programming algorithm that selects communities to find influential nodes within them. Then, they suggest an efficient algorithm called Parallelized Community-based algorithm (PCA) to parallel the influence propagation based on communities in order to improve the performance of their approach. Pozveh et al. (2017) suggest two approaches, C-SPIN (community aware SPIN) and C-SGA (community-aware SGA). C-SPIN and C-SGA are made up of three main phases: community detection, collection of candidate seeds and identification of influential nodes. Firstly, any accessible community detection algorithm based on the link structure can be applied, which is appropriate for the applied data sets taking into account the runtime. Secondly, the idea of blocked nodes is introduced into the minimal threshold and multiplication models in the candidate seed selection process. Then, based on the number and distance of other nodes activated or blocked by the nodes within communities, a centrality measure for ranking the nodes and selecting candidates is presented. This measure is also applied with some modifications to the linear threshold model, where blocked nodes does not exist. Finally, a heuristic approach to speed up the method of calculating the spread of influence is presented in the identification phase of the target set.

In order to identify influential nodes, Kitsak et al. (2010) proposed k-shell decomposition, and they discover that the influential nodes are those located in the core of network. Therefore, this decomposition affects multiple nodes in the same $k-shell$. Nonetheless, this index's high computational overhead makes it impractical to evaluate complex networks. Basaras, Katsaros, and Tassiulas (2013) proposed an alternative metric, the μ-power community index, which is a combination of coreness and betweenness centrality, μ-PCI is determined locally and therefore suitable for any type of network that ignores its size or dynamicity.

Ren et al. (2016) proposed a new method to classify influential nodes based on edge weight in a complex network, called evidential k-shell centrality. Firstly, according to Jaccard similarity, an edge weighting method is proposed for building weighted networks. Secondly, the value of the modified evidential centrality is calculated by considering real degree distribution. Thirdly, the layer of nodes located in the network is seen as a feature to classify influential nodes by combining modified evidential centrality. Only unweighted networks are studied by the authors, while many surveys could be done to identify influential nodes in direct networks.

4.2. Approaches in snapshots networks

Social networks in the real world continue to evolve over time, new accounts are created and some users will establish or lose contacts, the number of users and relationships are constantly changing. Moreover, no measure can accurately predict the dynamic change of social networks for two reasons: first, the relationships between users are randomly established second, those relationships may evolve over time. Static algorithms can't track and deal with these structural evolutions.

Many approaches model a dynamic social network as a series of snapshots. This representation makes it possible to adequately track the evolution of the network structure and then effectively identify the influential nodes. However, the analytical complexity also increases if the model's expressiveness is increased. When using network snapshots to perform time-conscious exploration tasks, two problems must be solved: (1) how to track the multiple evolutions of influential nodes when the network structure changes over time and (2) how to aggregate the results obtained in a previous snapshot with the results of subsequent ones.

4.2.1. Incremental approaches

Several real problems in many domains can be represented by dynamic networks with vertices representing social entities and edges representing interactions between entities at different times. A used representation to model social networks that evolve over time is the snapshot model, in which the network evolution is observed at different timestamps. An important problem under this model is the influential nodes detection where many approaches are proposed.

Zhuang, Sun, Tang, Zhang, and Sun (2013) examine the issue of influence maximization in dynamic networks where the graph evolution can only be observed by choosing some nodes (Fig. 10) from a timestamp to another. They contend that the evolution of the online social network cannot be completely tracked and identify a probing technique so that the actual process of influence propagation can be better identified with probing nodes. This paper's basic idea is to design a probing strategy so that the maximization solution obtained at the timestamp t based on the observed network can trigger the maximum influence propagation on the real network at t. A simple probing strategy is used to

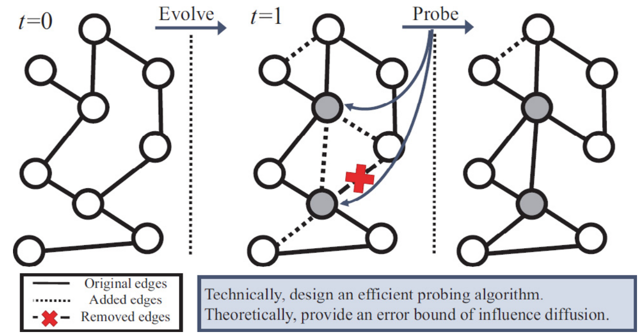

Fig. 10. An example of probing influence diffusion in a dynamic social network.

randomly select nodes of equal probability and probe their connection changes. However, because it treats all nodes equally, it is unlikely to minimize the loss of influence maximization performance. In Michalski, Kajdanowicz, Bródka, and Kazienko (2014), the objective was to confront the static context of seeding against a dynamic one. In the static approach, the social network used to identify influential nodes groups all past data into a single social network. In this paper, a new idea studied the temporal context of social networks that enables, in turn, to respect network dynamics detected in the past and make use of it for future identification of influential nodes in dynamic social networks. This concept is based on the computation of a set of structural measures for a historical series of networks, taking into account the sequential nature of time. As a result, the most recent network snapshots influence the final node ranking more than the older ones. This time-dependent ranking in a sense encapsulates the past dynamics of the social network. Liu, Li, Liao, Wang, and Wu (2012) used a static algorithm to identify the top-K nodes while the network structure is modified. The proposed algorithm SEMCE works as follows: First, the number of child nodes required for estimation is predicted according to the social network's power-law exponent. Then, based on the Monte Carlo estimation, several iterative steps are used to randomly analyze the nodes until finally the specified accuracy requirement is achieved. This method has some drawbacks that cannot be overlooked: (1) when the network grows in size, the runtime of a static approach can be very long (2) once the network topology has grown, it has to adjust the propagation control for all nodes, which allows very high costs.

In Chen, Song, He, and Xie (2015), the Upper Bound Interchange greedy algorithm (UBI) is proposed to track influential nodes, the nodes that maximize the influence in G^{t+1} are found based on those found in G^t. The UBI algorithm's objective is to identify the important nodes based on those previously found, rather than found them from an empty set. The authors apply the Interchange Heuristic as a method to replace the nodes found in the previous timestamp. Starting from an arbitrary set $S \subseteq V$, the Interchange Heuristic means finding a subset $S\prime \subseteq V$ that differs from S by one node, and has the same cardinality. Authors have shown that applying the exchange heuristic to the monotone submodular function until no further improvement is possible contributes to a solution with approximation guarantee 1/2. The growth of the network structure was not taken into consideration from which the similarity within the network structure leads to a comparable set of nodes that maximize the influence. Subsequently, they suggested the UBI+ algorithm that would achieve a better influence maximization. The researchers have extended their UBI algorithm to control influential nodes under the other diffusion model in dynamic networks. Their algorithm identifies a set of important nodes where the diffusion mechanism is in a static snapshot graph rather than in a dynamic network.

Liu et al. (2017) introduced an incremental approach described in Fig. 11, IncInf, to accurately locate the active nodes in changing social networks based on past data. Changes in the influence of nodes are visualized in IncInf by locating the evolution of topology in local communities only. A pruning strategy is also proposed to evaluate nodes with high degrees to locate the search space. First, by adopting the idea of localization, the authors design an effective approach for quantitatively analyzing the influence propagation changes from the network topology evolution. For the trade-off between efficiency and effectiveness an adjustable parameter is provided. Second, a pruning strategy is proposed that may effectively reduce the search space to nodes with only high degrees of scalability based on changes in the influence propagation and previous top-k nodes. An incremental DIM algorithm is suggested in Wang et al. (2017) to classify the powerful Top-K nodes in complex social networks. Instead of counting from scratch the influential nodes, the identification of those nodes is based on previous information. This approach is made up of two phases: initial identification of seeds and then the update strategy of seeds. DIM algorithm re-evaluates only the influence propagation of nodes affected by network change, nodes that have a slight change in the influence propagation still have little chance of being starting nodes. To further reduce the search space of influential nodes, the authors propose two pruning strategies for the DIM algorithm: the Least Influence Increment Pruning Strategy (LIIP) and the Degree Pruning Strategy (DP). In Hafiene, Karoui, and Romdhane (2019), the authors proposed SNDUpdate algorithm to detect the set of influential nodes that maximizes the influence within each snapshot graph G^t with varying the number of edges over each timestamp t. SNDUpdate uses an update function (Update b) to change the number of edges over each timestamp t, where b indicates the number of edges in each snapshot graph and then generates a set of snapshot graphs from networks by changing the number of edges with a constant time difference Δt.

The incremental approaches follow the principle of detecting influential nodes in G^{t+1} based on those detected in G^t.

Advantages: The major advantages of the incremental approaches are to improve the influence coverage and to incrementally detect influential nodes, without explicitly running the complete influence maximization algorithm at every snapshot.

Drawbacks: The major problem of incremental approaches is to conserve all previous information and then not observe the evolution of influential nodes between two snapshots. If the diffusion process starts with a wrong choice of seed nodes it ends with bad results.

4.2.2. Sliding window approaches

In this classification, the authors used a snapshot based time window representation: they divided the whole time interval of the temporal network into time windows, and for each window, they construct a static snapshot by aggregating all graphs from a sliding window of past time intervals.

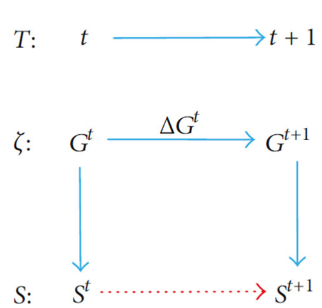

Fig. 11. IncInf: an incremental approach.

In Wang, Chakrabarti, Sivakoff, and Parthasarathy (2017), the authors introduced a three phases approach for identifying change points in dynamic social networks below the snapshot model. The authors modeled the evolution of the network as a Markov process of first order and thus the algorithm accounts for the temporal dependence while calculating the dissimilarity between the snapshots.

To identify important nodes in dynamic social networks based on the identification of such nodes in temporal communities that constitute the dynamic networks (Wang, Dai, Jiao, & Wang, 2016) proposed an algorithm that works as follows: First, to track influential nodes in each timestamp, they applied the degree corrected stochastic block (DCBM) (Karrer & Newman, 2011) in each snapshot. To further control the probability of edges between nodes, DCBM introduced the degree of nodes into the objective function. Second, they developed the extended Jaccard's coefficient to capture evolutionary relationships between communities at two consecutive time intervals. Third, they identified nodes that have long lived in a community to compose the set of influential nodes that initiate the spread of influence. Finally, to detect the influential nodes from those identified in the previous snapshot the authors applied three centrality measures.

Different types of approaches have been proposed where communities identified at time t are related to a trade-off of the best communities detected between t and the history of found communities. There are two main categories of such algorithms: those who update the communities existing at $t-1$ according to changes in the network between t and $t-1$, thus define a quality function as a deal between a quality function for communities at t (e.g Modularity) and a similarity between partitions at t and at $t-1$ (e.g Normalised Mutual Information NMI). The process usually follows three steps: (1) Detecting static communities on the first snapshot, (2) Using the network at $t+1$ and communities defined at t to detect communities at $t+1$ and (3) applying an algorithm to identify influential nodes in each snapshot.

Wei and Carley (2015) suggested two classes of dynamic measures to evaluate agents' temporal evolution in persistence and emergence. Persistence is a collection of measures to estimate the degree where an agent maintains his level of activity over time. Emergence is a class of metrics to evaluate changes in network activity between two adjacent timestamps. Every one of these dynamic measures relies on a particular model of temporal aggregation, that combines a series of temporal network activities into single numerical representations. The network activities are calculated by time stamp using static network metrics such as centrality of degree, centrality of authority or coefficients of clustering. They are grouped by one of the models of temporal aggregation and form only a Persistent or an Emergency measure. There are many drawbacks associated with this work. It is based on human actions' assumption of consistency. While their evaluation results support this issue, irregular human behavior may in some cases invalidate the basic model assumption.

The authors propose an efficient method in Wang et al. (2019) to identify influential nodes in dynamic networks by exploiting local detection and updating strategy. The basic strategy of the proposed local detection and updating approach is to identify the altered nodes in complex networks locally and to update the influence metrics of the altered nodes locally, without the need to quantify the influence of all nodes globally.

Firstly, they construct a dynamic evolution model to create a series of dynamic networks evolutions by adding and/or removing nodes and/or edges. Secondly, to calculate the effect of nodes in dynamic networks, they choose two metrics: (1) the degree centrality and (2) an improved metric derived from the Jaccard coefficient. Both of these metrics can be determined for each node based on local information. Finally, the authors develop a method for detecting and updating locally influential nodes in complex networks.

The sliding window approaches follow the principle of creating static snapshots by aggregating the activities of nodes and edges on different time windows.

Advantages: The major advantage of the sliding window approaches is studying both the structural and the dynamic properties of the network.

Drawbacks: The major problem of the sliding window approaches does not observe all the interactions between users when the sliding-window is too small. So, the user is considered as not influential, even though it has been influential for many years.

4.3. Approaches in dynamic networks

The below section discusses the review of various research carried out on the dynamic social network.

4.3.1. Interval approaches

Interval approaches are based on an interval graph, where interaction is represented as a quadruplet $(i, j, t, \delta t)$ which means that i is involved in contact with j from t to δt. In the interval graph, a link between two nodes exists for one or more periods. The link is defined not only by the two nodes that connect it, but also by a sequence of intervals (Fig. 12).

In Aggarwal, Lin, and Yu (2012), the authors proposed the first approach that resolving the problem of the authority for information spread in dynamic networks on the basis of temporary interactions. This paper studies both the problem of identifying a given model of influence node propagation and the influence propagation problem. They examined how to discover a set of nodes with the greatest influence over time. They modelled the influence propagation as a non-linear system, which is very different from trigger models such as the linear threshold model or the independent cascade model. In combination with forward temporal analysis the forward influence algorithm uses a greedy approach to determine the most influential nodes in the network. In practice, the forward influence algorithm is very slow, since it requires to estimate the spread of influence from different starting nodes on several times. Thus, the authors build faster approximations that use backward analysis to determine estimated influential nodes. The algorithm proposed in Aggarwal et al. (2012) is heuristic and the results produced do not accompany any proven quality guarantee. Under evolving networks, only a few papers concentrate on the influence maximization problem. Aggarwal et al. (2012) works on detecting a set of seeds within time t, optimizing the influence propagation at some $[t + \Delta]$ provided the dynamics of network evolution over time $[t, t + \Delta]$. The fundamental change is in Aggarwal et al. (2012), the process of diffusion is taking place under a dynamic network, while in other papers the authors find the influential nodes under a set of static snapshots taken from a dynamic social network.

The main objective of Bhowmick, Gueuning, Delvenne, Lambiotte, and Mitra (2019) is to identify a list of influential users in a social network based on temporal retweet sequence between two consecutive retweet events (receiving the tweet from her friend and retweeting it). Those users propagate the information to the full users in the social network. Given the sequence of inter-retweet time intervals and corresponding list of retweeting users, the goal of authors is to detect a seed that maximized the influence propagation at the end of the diffusion process. First, they set the ranked list of influential nodes from the temporal retweet sequence of cascades. This phase makes their algorithm more scalable. The authors then refine this list to diversify the exposed population through those powerful nodes. This method has the advantages of, on the one hand, not requiring the knowledge of the entire topological structure of the network, while, on the other hand, taking into account the heterogeneous behavior of users in a nontrivial manner.

The interval approaches follow the principle of choosing a set of nodes as active at a given interval time.

Advantages: The major advantage of the interval approaches is to detect the influential nodes in a dynamic network instead of a static snapshot.

Drawbacks: The major problem of interval approaches does not fully observe the network evolution between two consecutive time intervals.

4.3.2. A contact sequence approaches

In a sequence of contact graph, interaction is represented as a triple (i, j, t) where i and j are the interacting entities and t is the time when the relationship is activated (Fig. 13).

The IC model is extended in Chen, Lu, and Zhang (2012) to include the delay aspect of influence diffusion in social networks among persons, and take into account maximization of time-critical influence, where one wants to optimize the spread of influence within a specified time limit. The basic idea is to pick a budgeted subset to trigger influence, which might later optimize the influence propagation at timestamps of φ. Nevertheless, their research requires the full analysis of the dynamic network that is unlikely in the real world. As an example, after recently added relationships there are over 3 million, and every day 3 million are removed from the Weibo network. Providing a fully examined network is complicated at all times.

Ohsaka, Akiba, Yoshida, and Kawarabayashi (2016) has extended the IC model to include time information in the influence propagation process. The authors present the first full-dynamic index data structure in real-time designed to influence analysis on changing networks and provide a reachability-tree-based technique and a skipping method that can effectively reduce the time required for deletions and additions of edge/nodes, respectively.

Tong, Wu, Tang, and Du (2017) concentrated on the fact that the mechanism of diffusion in real dynamic social networks has many uncertain aspects and proposed a method of adaptive selection of seed nodes. With the above features, the influence maximization problem in social networks was studied. Through extending the classic IC model, they introduced the Dynamic Independent

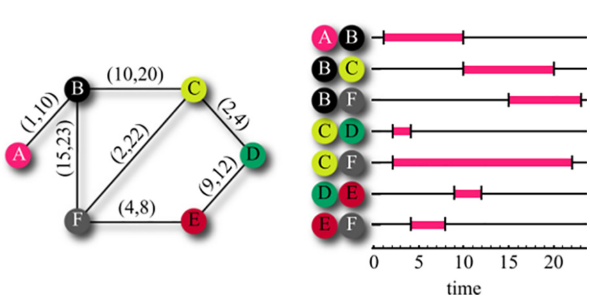

Fig. 12. Interval graph.

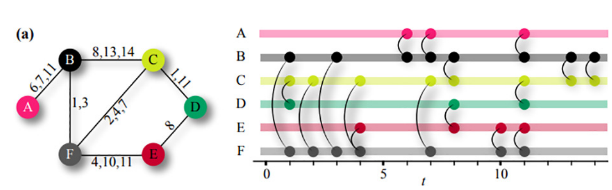

Fig. 13. A contact sequence graph.

Cascade (DIC) model; DIC can be able to control the dynamic features of real social networks. The authors introduce the concept of adaptive seeding strategy that represents a stochastic optimization framework used to narrow the candidate seed set before the seeding process. To design an adaptive seeding strategy they consider two problems: (1) how many budgets should they use in each seeding step and (2) which nodes to select. Throughout the classic IC model, a seed node will be triggered after selection and the relationship between two nodes will simply be explained by a fixed probability. While in the DIC model seed nodes cannot be activated with a certain probability, the probability of propagation between two nodes follows a certain distribution that explains topology changes in the social network.

In Yalavarthi and Khan (2018), the authors proposed a novel N-Family framework for dynamic influence maximization, that can be adapted to many influence maximization algorithms and several influence diffusion models. They developed a generalized local updating framework for efficiently adjusting the top-k influencers in an evolving network. Their method iteratively identified only the affected seed nodes due to dynamic updates in the influence graph, and then replaced them with more suitable ones. It can be adapted to many information diffusion models and influence maximization features. An update might affect the spread of influence of several nodes in the graph. However, compared to a distant node, the nearby nodes would be heavily impacted. The authors develop a threshold (θ)-based approach to classify the affected regions, this method is compatible with the MIA model. The main advantage between this approach and others that it needs to identify only a limited number of new seed nodes, based on affected regions due to updates. Other approaches need to identify all seed nodes from ground using modified sketches.

As a stream of edge weight updates, the authors in Yang, Wang, Pei, and Chen (2017) modeled a changing network. This model supports several network evolutions, including insertions of edges and nodes, deletions and modifications of weighted graphs. Under the popularly adopted linear threshold model and independent cascade model, they proposed two extremely important versions of the problem: find nodes with user-specified threshold influences and find the most important top-k nodes. Their key idea is to use polling-based approaches and hold a selection of random RR sets so they can estimate node influence with provable value guarantees. The proposed algorithm can increase the number of RR sets with changes in the network. They designed methods for evaluating the correct sample sizes for the two versions of the maximization problem. Thus, they can ensure good performance and be spatial and time-efficient at the same time. Zhao, Shang, Wang, Lui, and Zhang (2019) studied the problem of identifying influential nodes in a dynamic social graph where nodes and edges are in evolution. They proposed the Dynamic Time Decay Interaction Network (TDN) model that uses a time decay mechanism that captures the addition and deletion of evolving nodes and edges. This mechanism works as follows: an edge is added to the TDN at the time of its creation. The edge survives in the network for a time and then expires, when it is no longer alive, it will be removed from the network. If all edges attached to a node expire, then the node will be removed from the network. Fig. 14 shows an example of a TDN evolving process from time t to $t+1$.

The Chen et al. (2015) and Aggarwal et al. (2012) approaches are heuristic and have no proven quality. Although Ohsaka et al. (2016) authors claim that their algorithm has theoretical guarantees, a key parameter is described in their experiments, making the error rate even higher than 1. The justification the algorithm cannot be performed with little error rate in Ohsaka et al. (2016) is because the constant factor in its complexity is too high to be feasible to use it. Also, the influence model believed in Chen et al. (2015) and Ohsaka et al. (2016) is the Independent Cascade

$e \triangleq (u, v, \tau)$	$l_\tau(e)$
$e_1 = (u_1, u_2, t)$	1
$e_2 = (u_1, u_3, t)$	1
$e_3 = (u_1, u_4, t)$	2
$e_4 = (u_5, u_3, t)$	3
$e_5 = (u_6, u_4, t)$	1
$e_6 = (u_6, u_7, t)$	1
$e_7 = (u_5, u_2, t+1)$	1
$e_8 = (u_7, u_4, t+1)$	2
$e_9 = (u_7, u_6, t+1)$	3

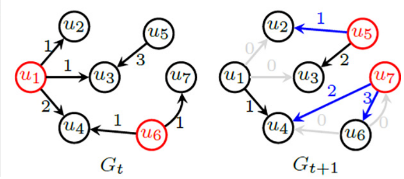

Fig. 14. TDN model.

model. The researchers looked at the Linear Threshold model as well as the Independent Cascade model in Yang et al. (2017). Ohsaka et al. (2016) and Yang et al. (2017) are the first to discuss the influence computation with demonstrable performance guarantees in complex networks. In Ohsaka et al. (2016), however, how to determine the retained RR sets is necessary to ensure the high quality of the extracted seed sets. Yang et al. (2017) offered solutions for finding influential nodes that exceeded the T threshold and detecting the $top - k$ influential individuals. Their strategies ensure that the recall is 100% with high probability and that the smallest effect of the individuals returned does not deviate from the true threshold by an absolute value ϵn. A major drawback of Yang et al. (2017) is that the authors need to set the parameters carefully in order to produce meaningful results. If the threshold T is too high, nothing can be returned by Yang et al. (2017). If the ϵ parameter, which regulates the absolute tolerable error in the top-k function, is not set correctly, ϵn can be even greater than the influence of the most influential k persons, and Yang et al. (2017) will return all the vertices that may be irrelevant. Therefore, to properly run algorithms in Yang et al. (2017), some previous knowledge of the data is needed, which is sometimes not simple to get.

The researchers have systematically discussed the two critical tasks of tracking top-k influential nodes in Yang, Wang, Jin, Pei, and Chen (2018): (1) tracking the most important top-k nodes and (2) supporting effective requests for the influence maximization problem. Their goal for both tasks is to monitor the error incurred in their algorithm (Yang et al., 2017) as a multiplicative variable to real data, so that they can still obtain meaningful results by setting a relative error threshold, even without prior knowledge of the data.

The contact sequence approaches follow the principle of incorporating the temporal contact between nodes and edges into dynamic social networks.

Advantages: The major advantages of the contact sequence approaches are to estimate the importance of nodes and edges in a contact sequence graph by applying metrics and to guarantee a good quality of seed sets extracted.

Drawbacks: The major problem with many contact sequence approaches is to locate all seed nodes from the ground which is very expensive, even for moderate size networks.

5. Comparison

A technical comparison of the different approaches is highlighted in Table 2, where details of the algorithms can be found in Sections 4. In the first column of Table 2, we classify approaches based on network models: static networks, snapshots networks and dynamic networks. In the second column, we define subcategories for each of these models, corresponding to the different techniques used to study the influence maximization problem. From column 4 to column 8, we indicate whether the compared algorithms support different diffusion models (✔for support, × for not support). In columns 9, we give the techniques of the algorithms respectively. In column 10, we describe the different data sets used in each approach to identify the influential nodes.

Table 2
A technical comparison of influence maximization approaches.

Category		Methods	IC	LT	TR	WC	SIR	Techniques	Data sets
Static	Greedy	Wang and Yang, (2009)	✓	✓	×	✓	×	RanCas(S) function	NetHEPT and NetPHY
		Leskovec et al. (2007)	✓	✓	✓	×	×	Submodular function	Blog networks and water networks
		Zhou et al. (2013)	✓	×	×	×	×	Upper bound function	ca-GrQc, ca-HepPh, Digger and email-Enron
	Heuristic	Narayanam and Narahari (2010)	×	✓	×	×	×	Shapley value function	sparse random graph, scale-free graph, Political Books, Jazz, Celegans, NIPS, Netscience and HEP
		Bae and Kim (2014)	×	×	×	×	✓	k-shell decomposition method	Twelve real networks and artificial networks
		Wang et al. (2015)	×	×	×	✓	×	Price-Performance Ratio and Influential Power	Political blogs and Neural network
	Hybrid	Cao et al. (2011)	✓	✓	×	×	×	Growth function	NetHEPT and NetPHY
		Ren et al. (2016)	×	×	×	×	✓	Evidential k-Shell Centrality	Zachary, Jazz, Dolphin and Email
		Pozveh et al. (2017)	×	✓	×	×	×	The community structure and influence based-closeness centrality measure	Political Books, Celegans, Netscience co-authorships, Zachary's Karate Club and Adjacency of nouns and Delicious
		Kitsak et al. (2010)	×	×	×	×	✓	K-shell decomposition and the imprecision function	Email, LiveJournal, Adult IMDB and CNI networks
		Song et al. (2014)	✓	×	×	×	×	Divide-and-conquer strategy with parallel computing mechanism	Four Mobile Social Networks
Snapshots	Incremental	Liu et al. (2012)	✓	×	×	×	×	Monte Carlo estimation strategy	NetHEPT, NetPHY, Enron, Epinions and Amazon
		Zhuang et al. (2013)	✓	×	×	×	×	Probing strategy	Synthetic, Twitter and Coauthor
		Chen et al. (2015)	×	×	×	✓	×	Interchange Heuristic strategy	Mobile, HepPh and HepTh
		Liu et al. (2017)	✓	×	×	×	×	Pruning strategy	Facebook, NetHEPT, and Flickr
		Wang et al. (2017)	×	✓	×	×	×	Pruning strategy	NetHEPT, Facebook, Flixster and Flickr
	Sliding window	Wei and Carley (2015)	Temporal aggregation models					Persistence and Emergence metrics	Newcomb, Enron, DBLP and Arab Spring dataset
		Wang et al. (2016)	DCBM model					Independent clustering, the extended Jaccard's coefficient and centrality metrics	Real and synthetic networks and LFR benchmarks graphs
		Wang et al. (2019)	Dynamic evolution model					The strategy of Local Detection and Updating	The Barabasi-Albert (BA) scale-free network, the Watts-Strogatz (WS) small-world network, and the Erdos-Rényi (ER) random network.
Dynamic	Interval	Aggarwal et al. (2012)	Influence pattern model					Globally optimized forward trace approach and a locally optimized backward approach	DBLP and Arnetminer Citation
		Bhowmick et al. (2019)	Temporal retweet model					Twitter Specific Metrics	Algeria, Egypt, and Nepal
	A contact sequence	Ohsaka et al. (2016)	✓	×	✓	✓	×	Reverse reachable sketch	Digg, Enron, Epinions, Facebook, DBLP, YouTube and Flickr
		Tong et al. (2017)	DIC model					Adaptive Seeding Strategy	Hep, Wiki and PL
		Yalavarthi and Khan (2018)	MIA model and IC					Localized batch updates technique	Digg, Slashdot, Epinions and Flickr

Static networks can be used to model real-world problems which neglected the evolutionary nature of the world in which we live. In reality, social interactions evolve over time, hence the need to study snapshot and dynamic networks. In this survey, we don't compare static networks with snapshot and dynamic ones, because by definition, temporary information cannot be used directly in static networks, thus reducing dynamic problems to static problems.

The important issues defined in static environments have been reformulated and temporal approaches capable of solving them have been proposed. In this section, an important issue to be addressed is the relation between snapshot and temporal networks. The choice to use dynamic networks instead of snapshot networks (or vice versa) can have a significant implication on aspects related to data storage and temporal processing of the network evolutions.

When we try to capture the network evolution in a specific moment, the snapshot model is the most adopted. At each snapshot, we checked the network and based on the information observed in it we continue the network evolution. Contrary, if more accurate temporal information is available with infinite precision of timestamps than the dynamic networks can be the best considered. Snapshot networks are composed of aggregated times. Thus, on this type of networks the typical complexity of the evolution processes based on the selected level of aggregation, and the dimensions of the generated snapshots (node/edge number).

A high number of vertices and edges is observed in the networks, as many variations in the number of nodes and edges are registered during the time window. This number does not show great variations in time for a given timestamp. However, it remains very limited when we considered the fully dynamic networks. When studying the different proposed approaches in the snapshot networks, we observed a problem which is the absence of strong evolutions in the number of influential nodes among two consecutive timestamps. In the snapshot networks, when aggregating data over time windows not only the network size increase but also the network become denser, thus the number of active relations growing faster than the number of active nodes. In dynamic networks, the evolution processes focus mainly on the number of network changes, not on the granularity or the total number of nodes.

6. Challenges and future directions

When analyzing the social influence and with the increasing development of social networks, major challenges have been posed. Although many researchers have proposed solutions to some problems related to the influence analysis and maximization problem, many others are not sufficiently studied and need to be explored. This survey demonstrates challenges and future directions with a focus on maximizing the influence in dynamic social networks.

6.1. Challenges

Nowadays, the influence analysis in social networks represents an interesting area of research. Since this field is still recent, we have observed some problems that have not been covered yet by the researchers.

1. Online social networks are heterogeneous and contain different types of edges which represent relation between two nodes. In social networks, the influence is propagated over networks along social links, which produces different types of influence. For instance, Twitter is composed of users and microblogs where links between users are directed from users to their followers. Digg includes users and URLs of websites where links between users are not directed. The citation network contained authors and published articles. Therefore, it is hard to know how to extract information from directed and undirected links in heterogeneous social networks.
2. In real-world users can belong to many social networks. However, the existing influence diffusion models are adopted for a specific social network. For example, the hybrid factor non-negative matrix factorization model is proposed to predict social influence in Twitter. Therefore, it is difficult to collect separately the information propagated within a specific social network.
3. Social network is constantly evolving, being able to react to these changes quickly is very crucial. Further, the social relationship between any two users and the network topology evolve over time. This uncertainty of social graph inspires us to study a different kind of social influence analysis. However, research in this domain is still limited, and much work needs to be completed to design evolution models that can track both the structure of the network structure and the social relationship between users.
4. The social web contains a variety of social media platforms, which cover a wide range of user activities. With the fact that a single user has multiple accounts on different social networks, it has become increasingly important to link distributed user profiles belonging to the same user from multiple sources, which can help to understand how different types of Social interactions between users can influence the evolution of the nodes of influence in social networks.
5. Researchers have a duty to report on the findings of their research and they cannot do so if the data they collect from social networks is confidential. What researchers can do is to guarantee that they do not reveal identifiable information about users and attempt to protect the research users' privacy by various processes designed to anonymize them.

6.2. Future directions

We have studied in the previous section the problems presented in the field of influence analysis on social networks. In this section, we propose some future recommendations for more research in this field.

1. The current models for influence evaluation are limited, they consider either a model that contains the characteristics of users and the evolution of the relationship between nodes over time or a model that containing information about the evolution of the network structure. In the real-world online social networks, such as Twitter or Facebook, both network structure and relationship between users change over time. Therefore, we propose to describe hybrid approaches that combine the benefits of community detection problem with influential node detection algorithms to improve the efficiency and effectiveness of social influence analysis.
2. Information used in existing works is collected from heterogeneous social networks. It is necessary to integrate both textual information and link information collecting from various networks in a generative modelling framework to extract user interests which are represented as mixtures of topics and direct influence between users simultaneously.
3. The influence maximization problem is a complex problem. To correctly represent and model this problem, several fields can be combined, like graph theory, deep learning, and multi-agent systems.
4. Existing approaches that studied the influence maximization problem suffer from a scalability problem. With the increasing size of social networks and the evolution of edges and nodes over time, it is difficult to track them in a complex social network. Thus, it is important to propose several models to provide the scalability and the effectiveness of the network size and structure in dynamic social networks.

7. Conclusion

In this survey, we provide an overview of the current status of influential node detection problem in social networks and their future directions. We reviewed the existing literature in this field and categorized it into three classifications that are based on network models: static networks, snapshots networks and dynamic networks. This survey demonstrates that by proposing different algorithms, methodologies, and frameworks, numerous research works have attempted to solve the influential nodes detection problem.

In conclusion, we identified some open issues in influential nodes detection. First, the majority of the current field literature defines influential nodes for a single topic, one might be interested to investigate how influential nodes vary substantially across various topics (such as economics, sports, media, etc.). Second, many researchers like to generalize their algorithms to track influential nodes under other diffusion models such as the Continuous Time Diffusion Model. Third, parallelizing many existing methods in large distributed systems is an interesting future direction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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